TEST YOUR UNDERSTANDING

For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. Three of the following are evidence that charophytes are the closest algal relatives of plants. Select the exception.
(A) similar sperm structure
(B) the presence of chloroplasts
(C) similarities in cell wall formation during cell division
(D) genetic similarities in chloroplasts

2. Which of the following characteristics of plants is absent in their closest relatives, the charophyte algae?
(A) chlorophyll *b*
(B) cellulose in cell walls
(C) sexual reproduction
(D) alternation of multicellular generations

3. In plants, which of the following are produced by meiosis?
(A) haploid gametes
(B) diploid gametes
(C) haploid spores
(D) diploid spores

4. Microphylls are found in which plant group?
(A) lycophytes
(B) liverworts
(C) ferns
(D) hornworts

Levels 3-4: Applying/Analyzing

5. Suppose an efficient conducting system evolved in a moss that could transport water and other materials as high as a tall tree. Which of the following statements about "trees" of such a species would be true?
(A) Spore dispersal distances would probably decrease.
(B) Females could produce only one archegonium.
(C) Unless its body parts were strengthened, such a "tree" would probably flop over.
(D) Individuals would probably compete less effectively for access to light.

6. Identify each of the following structures as haploid or diploid.
a. sporophyte
b. spore
c. gametophyte
d. zygote

7. **EVOLUTION CONNECTION • DRAW IT** Draw a phylogenetic tree that represents our current understanding of evolutionary relationships between a moss, a gymnosperm, a lycophyte, and a fern. Use a charophyte alga as the outgroup. (See Figure 26.5 to review phylogenetic trees.) Label each branch point of the phylogeny with at least one derived character unique to the clade descended from the common ancestor represented by the branch point.

Levels 5-6: Evaluating/Creating

8. **SCIENTIFIC INQUIRY • INTERPRET THE DATA** The feather moss *Pleurozium schreberi* harbors species of symbiotic nitrogen-fixing bacteria. Scientists studying this moss in northern forests found that the percentage of the ground surface "covered" by the moss increased from about 5% in forests that burned 35 to 41 years ago to about 70% in forests that burned 170 or more years ago. From mosses growing in these forests, they also obtained the following data on nitrogen fixation:

Age (years after fire)	N fixation rate [kg N/(ha · yr)]
35	0.001
41	0.005
78	0.08
101	0.3
124	0.9
170	2.0
220	1.3
244	2.1
270	1.6
300	3.0
355	2.3

Data from O. Zackrisson et al., Nitrogen fixation increases with successional age in boreal forests, *Ecology* 85:3327–3334 (2006).

(a) Use the data to draw a line graph, with age on the x-axis and the nitrogen fixation rate on the y-axis.
(b) Along with the nitrogen added by nitrogen fixation, about 1 kg of nitrogen per hectare per year is deposited into northern forests from the atmosphere as rain and small particles. Evaluate the extent to which *Pleurozium* affects nitrogen availability in northern forests of different ages.

9. **WRITE ABOUT A THEME: INTERACTIONS** Giant lycophyte trees had microphylls, whereas ferns and seed plants have megaphylls. Write a short essay (100–150 words) describing how a forest of lycophyte trees may have differed from a forest of large ferns or seed plants. In your answer, consider how the type of forest may have affected interactions among small plants growing beneath the tall ones.

10. **SYNTHESIZE YOUR KNOWLEDGE**

These stomata are from the leaf of a common horsetail. Describe how stomata and other adaptations facilitated life on land and ultimately led to the formation of the first forests.

For selected answers, see Appendix A.